DRISHTI

AUTHORS

PRIYANKA CHIROTE 0827CS221201
PRIYAL WAGHMARE 0827CS221200
REETI SHRIMAL 0827CS221223
RIYA KAUSHAL 0827CS221230
MANDISHA MIRZA 0827CS233D14

AFFILIATIONS

ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESERCH

INTRODUCTION

Conventional surveillance systems rely on manual observation, making them inefficient and prone to human error. With advances in AI and computer vision, real-time automated surveillance can now detect, recognize, and track individuals with high precision.

Drishti is designed to intelligently monitor environments by integrating real-time human detection, headcounting, and facial recognition into a single automated framework, reducing manual effort and improving situational awareness.

OBJECTIVE

To develop an AI-powered surveillance system capable of:

- Real-time detection and counting of individuals.
- Identifying familiar faces through CNN-based recognition.
- Maintaining privacy by excluding unknown identities from permanent storage.
- Providing a secure and scalable dashboard for automated monitoring.

METHODOLOGY

The system follows an end-to-end modular process:

1. Data Loader: Collects diverse face profiles under varying angles and lighting.
2. Preprocessing: Normalizes and enhances facial images; crops faces via Haar/DNN detectors.
3. Model Training: A CNN extracts embeddings and learns discriminative features from the dataset.
4. Real-Time Detection: Live video frames are processed using AI models (e.g., YOLO/CNN).
5. Recognition: Matches detected faces against the trained database.
6. Database & UI: Recognized faces stored securely; unknowns tagged but not saved.
7. The system architecture includes a camera module, processing engine, secure database, and live dashboard for user interaction.

ANALYSIS

Testing was performed in classrooms and labs with moderate crowd sizes.

Observations:

- High accuracy in controlled lighting.
- Efficient tracking of multiple faces simultaneously.
- Simplified real-time monitoring via dashboard visualization.
- Limitations:
- Accuracy drops in low-light or partial-occlusion conditions.
- Scalability to multi-camera systems increases computational load.

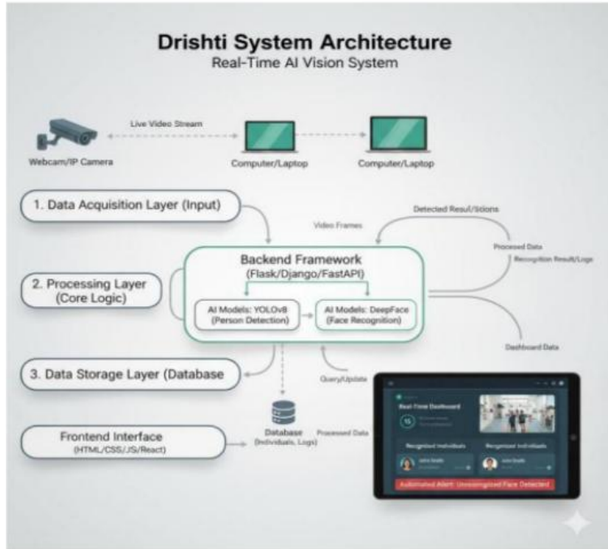

FIGURE 1.1 : ADVANCE SYSTEM ARCHITECTURE OF DISHTI MODEL

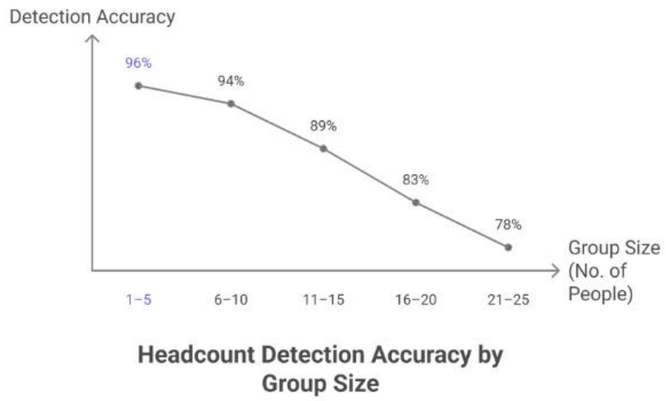

FIGURE 1.2 : HEAD COUNT ACCURACY BY GROUP SIZE

RESULTS/FINDINGS

- Achieved reliable recognition of trained individuals with high precision.
- Provided dynamic headcount statistics in real-time, even under varying crowd densities.
- Reduced manual monitoring requirements significantly through automation.
- Ensured privacy protection by excluding unidentified faces from database storage.
- Demonstrated smooth operation across different camera angles and moderate lighting variations.
- Produced consistent outputs during continuous surveillance sessions, confirming system stability.
- Maintained >90% accuracy under controlled conditions with minimal latency during recognition tasks.

CONCLUSION

The proposed Drishti system successfully bridges the gap between conventional manual surveillance and modern intelligent monitoring. By combining real-time headcounting, facial recognition, and automated data management, it provides a comprehensive framework capable of enhancing both security and efficiency in sensitive environments like classrooms, offices, and laboratories.

The system's modular architecture ensures flexibility and scalability for future expansion into multi-camera networks, IoT-based alerting, and large-scale deployments. Through the integration of deep learning models and secure database protocols, Drishti delivers actionable insights that enable informed decision-making while maintaining ethical standards of privacy and data protection.